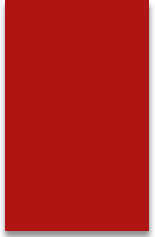


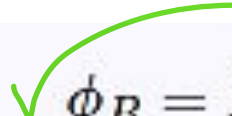


Electromagnetic Induction

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Magnetic flux is defined as the number of magnetic field lines passing through a given closed surface. It provides the measurement of the total magnetic field that passes through a given surface area.


$$\phi_B = B \cdot A = BA \cos \Theta$$

•The SI unit of magnetic flux is **Weber (Wb)**.

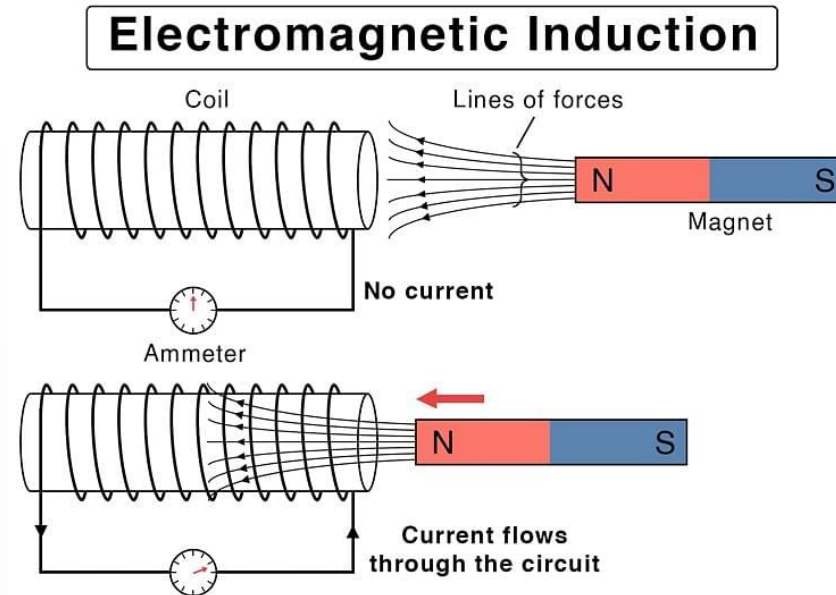
Faraday's Laws of Electromagnetic Induction

Faraday's law of electromagnetic induction (referred to as **Faraday's law**) is a basic law of electromagnetism that predicts how a magnetic field will interact with an electric circuit to produce an electromotive force (EMF). This phenomenon is known as electromagnetic induction. Faraday's Laws of Electromagnetic Induction consist of two laws. The first law describes the induction of emf in a conductor and the second law quantifies the emf produced in the conductor.

Faraday's law of induction explains the working principle of transformers, motors, generators, and inductors. The law is named after Michael Faraday, who performed an experiment with a magnet and a coil. During Faraday's experiment, he discovered how EMF is induced in a coil when the flux passing through the coil changes.

Faraday's First Law

Whenever a conductor is placed in a varying magnetic field, an electromotive force (EMF) is induced. If the conductor circuit is closed, a current is induced, which is called induced current.



Mentioned here are a few ways to change the magnetic field intensity in a closed loop:

- By rotating the coil relative to the magnet.
- By moving the coil into or out of the magnetic field.
- By changing the area of a coil placed in the magnetic field.
- By moving a magnet towards or away from the coil.

Faraday's Second Law

The induced emf is directly proportional to the rate of change in magnetic flux. As you will see in the next section (Len's law), the induced emf tends to oppose the flux change the minus sign indicating that opposition

$$\varepsilon = - \frac{d\phi}{dt}$$

If we change the magnetic flux through a coil of N turns, an induced emf appears

$$\varepsilon = - N \frac{d\phi}{dt}$$

1. A coil is placed in a transverse magnet field of 0.5 T. If this coil starts shrinking at a rate of 0.1 mm/sec, while its radius remains 5 cm, then what is the value of induced emf?
2. Calculate the magnitude of the induced emf when the magnet is thrust into a coil. The following information is given: the single loop coil has a radius of 5 cm, and the average value of the complex magnetic field component $B \cos \theta$ increases from 0.1 T to 0.5 T in 0.2 s.



Applications of Faraday's Law

- Faraday law is one of the most basic and important laws of electromagnetism. This law finds its application in most of the electrical machines, industries, and the medical field, etc.
- Power transformers function based on Faraday's law
 - The basic working principle of the electrical generator is Faraday's law of mutual induction.
 - The Induction cooker is the fastest way of cooking. It also works on the principle of mutual induction. When current flows through the coil of copper wire placed below a cooking container, it produces a changing magnetic field. This alternating or changing magnetic field induces an emf and hence the current in the conductive container, and we know that the flow of current always produces heat in it.
 - Electromagnetic Flow Meter is used to measure the velocity of certain fluids. When a magnetic field is applied to an electrically insulated pipe in which conducting fluids are flowing, then according to Faraday's law, an electromotive force is induced in it. This induced emf is proportional to the velocity of fluid flowing.
 - From bases of Electromagnetic theory, Faraday's idea of lines of force is used in well known Maxwell's equations. According to Faraday's law, change in magnetic field gives rise to change in electric field and the converse of this is used in Maxwell's equations.
 - It is also used in musical instruments like an electric guitar, electric violin, etc.

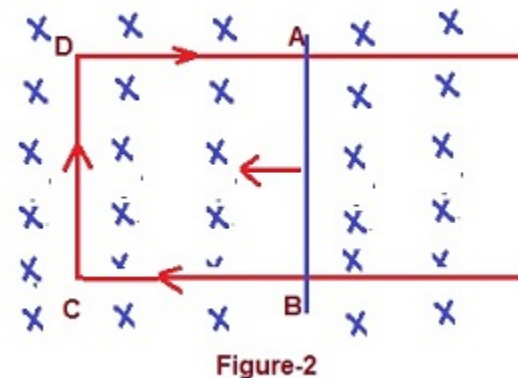
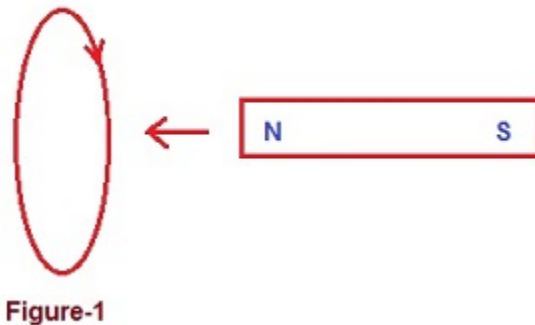
Lenz's Law

To obey the conservation of energy, the direction of the current induced via Lenz's law must create a magnetic field that opposes the magnetic field that created it in the first place.

*The direction of the induced current generated by the induced EMF is such that it opposes the cause producing it. In other words, it is defined as “**Effect Opposes the Cause**”*

Illustration of Lenz's Law

In figure (1) below, a magnet is brought toward a circular loop. The North Pole of magnet is facing the loop. As the magnet get closer to the loop, the magnetic field through the loop increases. Due to which, the flux of the magnetic field through the closed loop increases. Mind here that, the cause is increasing magnetic flux and effect is induced current. Therefore as per Lenz's Law, the effect i.e. induced current shall oppose the cause. Therefore, induced current will produce a magnetic flux which will weaken the original magnetic flux. The direction of original magnetic field is going into the plane of conducting loop. Therefore, the induced current shall produce magnetic field toward the magnet or away from the plane of conducting loop to oppose the cause. Hence, the direction of induced current is anticlockwise by Screw Rule.





Consider the figure (2). As shown in the figure, the wire AB is sliding toward left. Therefore, the area ABCD decreases with the slide of wire AB. Notice here that magnitude and direction of magnetic field is constant. But as the area ABCD is decreasing, the flux linkage $\Phi = BA$ will decrease. Therefore, from [Faraday's Law](#), a current will be induced in the circuit ABCD.

Now we will use Lenz's Law to find the direction of induced current. Here, the cause is decreasing magnetic flux and effect is [induced current](#). Effect should oppose the cause. Therefore, the direction of induced current shall be such that it strengthens the magnetic flux. The direction of original magnetic field is going into the plane of your device screen. Induced current should also produce magnetic field in the same direction which means the direction of induced current should be clockwise.

Inductors and Inductance

We found that a capacitor can be used to produce a desired electric field. We considered the parallel-plate arrangement as a basic type of capacitor. Similarly, an inductor can be used to produce a desired magnetic field. We shall consider a long solenoid as our basic type of inductor. If we establish a current i in the windings (turns) of the solenoid we are taking as our inductor, the current produces a magnetic flux B through the central region of the inductor. The inductance of the inductor

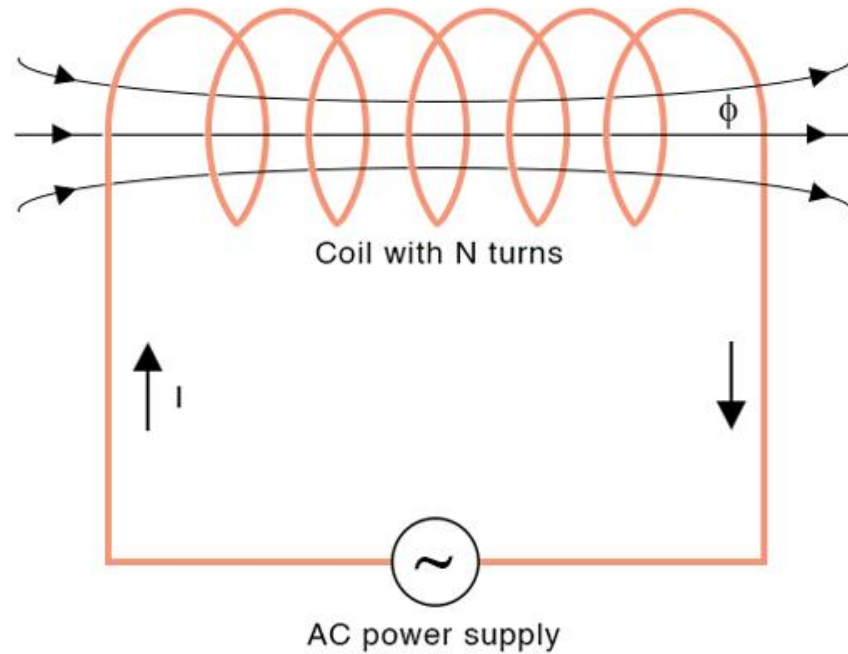
$$L = \frac{N\phi_B}{i}$$

The inductance L is thus a measure of the flux linkage produced by the inductor per unit of current. the SI unit of inductance is the tesla-square meter per ampere (Tm^2/A). We call this the Henry (H), after American physicist Joseph Henry.

Self Induced EMF : a self-induced emf appears whenever the current changes with time. The magnitude of the current has no influence on the magnitude of the induced emf; only the rate of change of the current counts

$$\varepsilon = -L \frac{di}{dt}$$

Inductance of a Solenoid

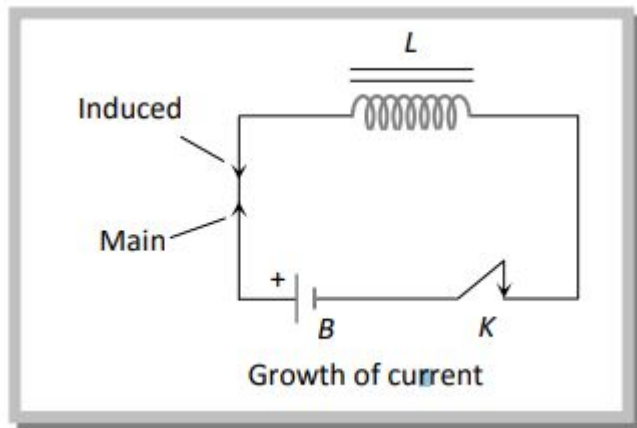


$$L = \mu_0 n^2 \pi R^2 l$$

Inductance—like capacitance—depends only on the geometry of the device.

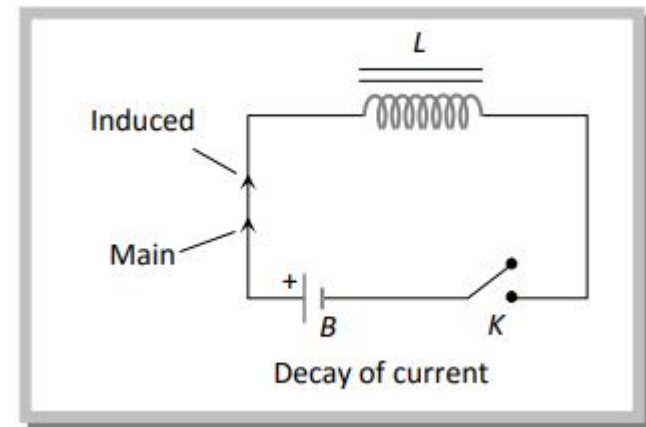
Growth and Decay of Current in LR-Circuit.

If a circuit containing a pure inductor L and a resistor R in series with a battery and a key then on closing the circuit current through the circuit rises exponentially and reaches up to a certain maximum value (steady state). If circuit is opened from its steady state condition then current through the circuit decreases exponentially.



The value of current at any instant of time t after closing the circuit (i.e. during the rising of current) is given by

$$i = i_0 \left[1 - e^{-\frac{R}{L}t} \right]$$

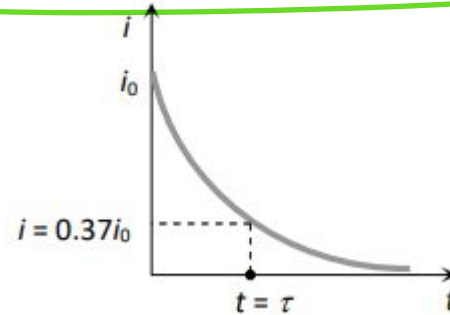
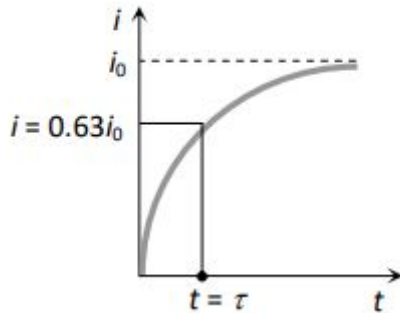


The value of current at any instant of time t after opening from the steady state condition (i.e. during the decaying of current) is given by

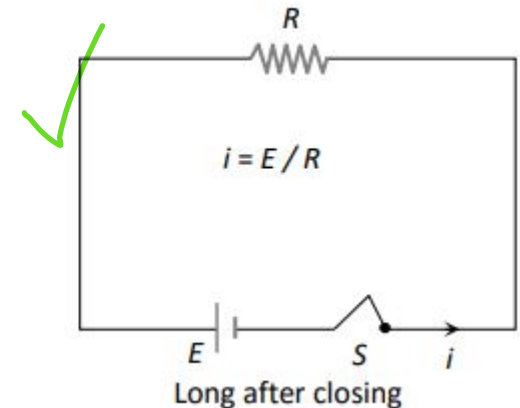
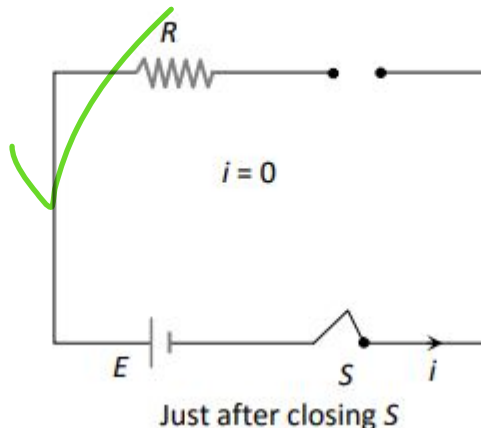
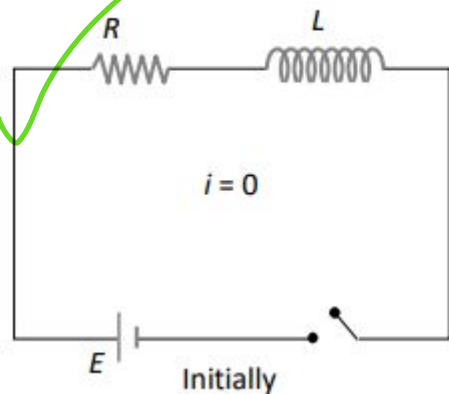
$$i = i_0 e^{-\frac{R}{L}t}$$

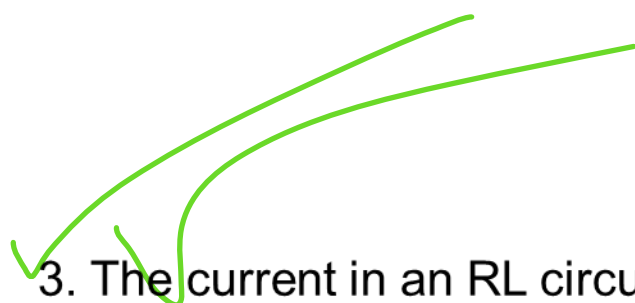
Inductive time constant (τ)

The time interval, during which the current in an inductive circuit rises to 63% of its maximum value at make, is defined as time constant or it is the time interval, during which the current after opening an inductive circuit falls to 37% of its maximum value.



The current in the circuit grows exponentially with time from 0 to the maximum value I (E/R). Just after closing the switch as $i = 0$, inductor act as open circuit i.e. broken wires and long after the switch has been closed as $i = i_0$, the inductor act as a short circuit i.e. a simple connecting wire.



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3. The current in an RL circuit builds up to one-third of its steady-state value in 5.00 s. Find the inductive time constant.
4. The current in an RL circuit drops from 1.0 A to 10 mA in the first second following the removal of the battery from the circuit. If L is 10 H, find the resistance R in the circuit.
5. A battery is connected to a series RL circuit at time $t = 0$. At what multiple of τ_L will the current be 0.100% less than its equilibrium value?

3 The steady state value of the current is also its maximum value, $\mathcal{E}/R$, which we denote as i_m . We are told that $i = i_m/3$ at $t_0 = 5.00$ s. Equation 30-41 becomes $i = i_m(1 - e^{-t_0/\tau_L})$, which leads to

$$\tau_L = -\frac{t_0}{\ln(1 - i/i_m)} = -\frac{5.00 \text{ s}}{\ln(1 - 1/3)} = 12.3 \text{ s}.$$

4 The current in the circuit is given by $i = i_0 e^{-t/\tau_L}$, where i_0 is the current at time $t = 0$ and τ_L is the inductive time constant (L/R). We solve for τ_L . Dividing by i_0 and taking the natural logarithm of both sides, we obtain

$$\ln\left(\frac{i}{i_0}\right) = -\frac{t}{\tau_L}.$$

This yields

$$\tau_L = -\frac{t}{\ln(i/i_0)} = -\frac{1.0 \text{ s}}{\ln((10 \times 10^{-3} \text{ A})/(1.0 \text{ A}))} = 0.217 \text{ s}.$$

Therefore, $R = L/\tau_L = 10 \text{ H}/0.217 \text{ s} = 46 \Omega$.

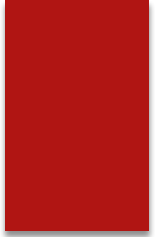


5. Starting with zero current at $t = 0$ (the moment the switch is closed) the current in the circuit increases according to

$$i = \frac{\mathcal{E}}{R} (1 - e^{-t/\tau_L}),$$

where $\tau_L = L/R$ is the inductive time constant and $\mathcal{E}$ is the battery emf. To calculate the time at which $i = 0.9990\mathcal{E}/R$, we solve for t :

$$0.9990 \frac{\mathcal{E}}{R} = \frac{\mathcal{E}}{R} (1 - e^{-t/\tau_L}) \Rightarrow \ln(0.0010) = -(t/\tau) \Rightarrow t/\tau_L = 6.91.$$



Mutual inductance

Mutual inductance is the main operating principle of generators, motors and transformers.

What Is Mutual Inductance?

When two coils are brought in proximity to each other, the magnetic field in one of the coils tends to link with the other. This further leads to the generation of voltage in the second coil. This property of a coil which affects or changes the current and voltage in a secondary coil is called mutual inductance.

- i_1 in Coil 1 increases
- More is B_1
- More will be field lines through Coil 2
- More flux through Coil 2

$$(\phi_2)_T \propto i_1$$

$$(\phi_2)_T = M i_1$$

$$M = (\phi_2)_T / i_1$$

Induced EMF in Coil 2

$$\varepsilon_2 = -\frac{d(\phi_2)_T}{dt} = -\frac{dM i_1}{dt} = -M \frac{d i_1}{dt}$$

